



General Physics II

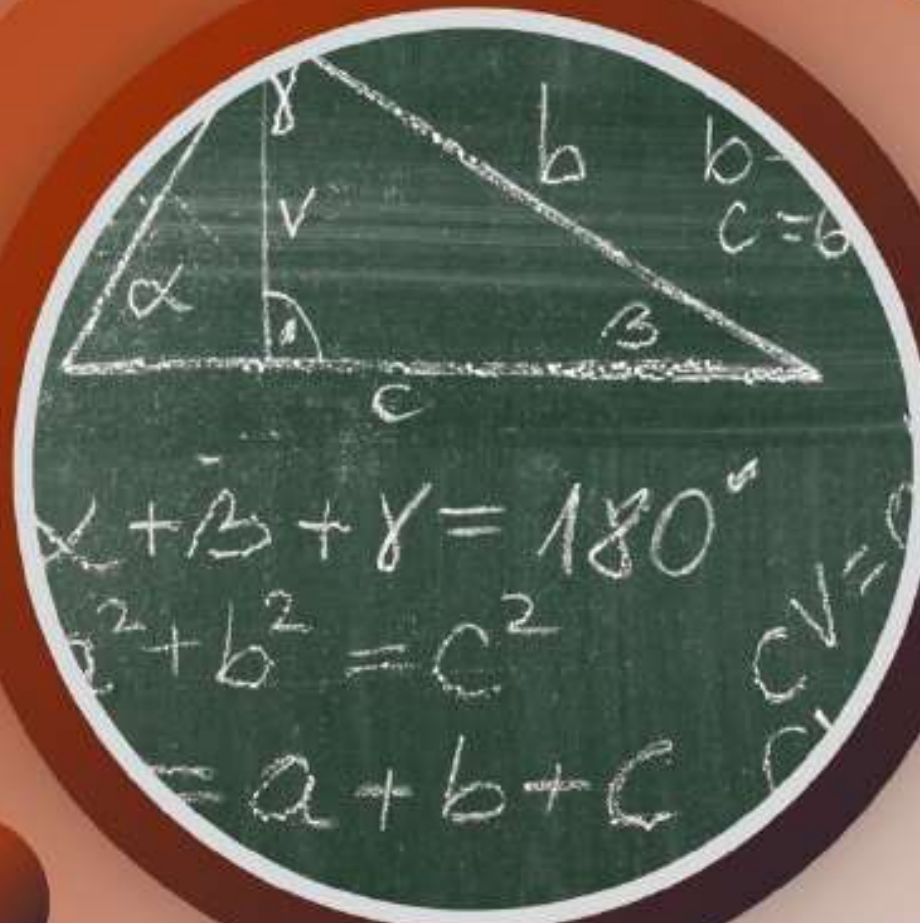
PHY121

College of Science and Technology



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MODULE 4: MAGNETISM & EM INDUCTION

UNIT 1: MAGNETIC FIELDS AND LORENTZ LAW

Magnetic fields

A magnetic field is a vector field that describes the magnetic influence on moving electric charges, electric currents, and magnetic materials. A moving charge in a magnetic field experiences a force perpendicular to both its velocity and the magnetic field. its symbol “B”. Unit

$$= \text{Tesla}, 1\text{T} = 1 \frac{\text{N.s}}{\text{C.m}}$$

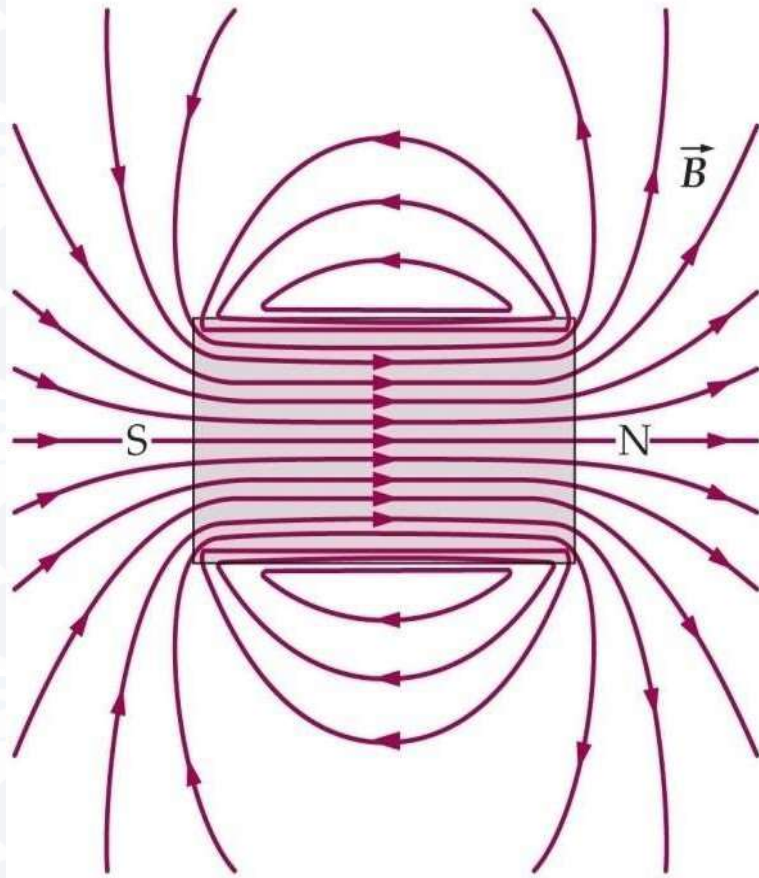
Properties

- Magnetic field lines form closed loops.
- Direction of field lines: from the North to the South pole outside a magnet.
- Field lines never intersect.
- Strength is denoted by the density of lines (closer = stronger field).

Sources of Magnetic Fields

- Moving charges (currents)
- Permanent magnets
- Changing electric fields (Maxwell's extension)

Magnetic field lines



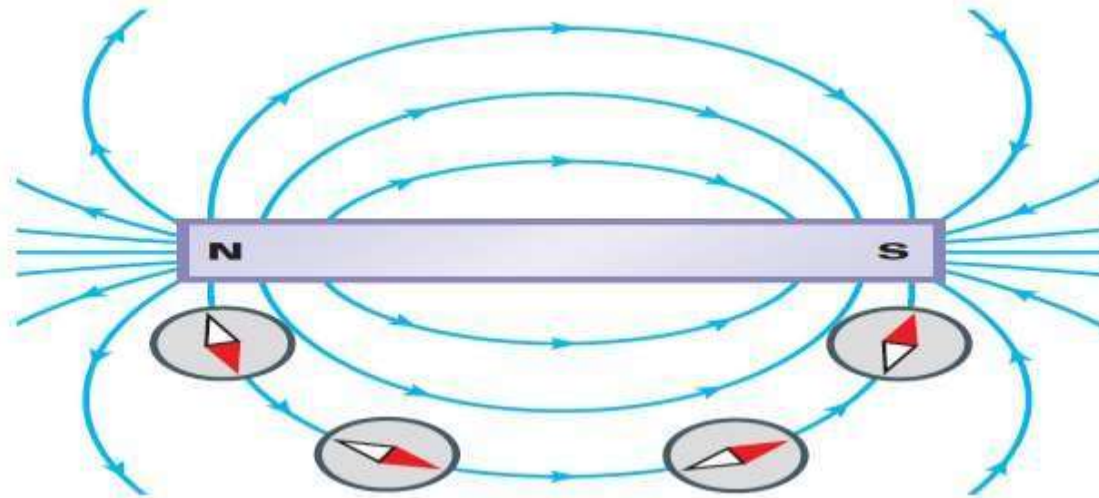
Magnetic field $\underline{B}$ can be represented by field lines, and as with electric field lines, the direction of the field is indicated by the direction of the field line, and the magnitude of the field is indicated by the density of lines.

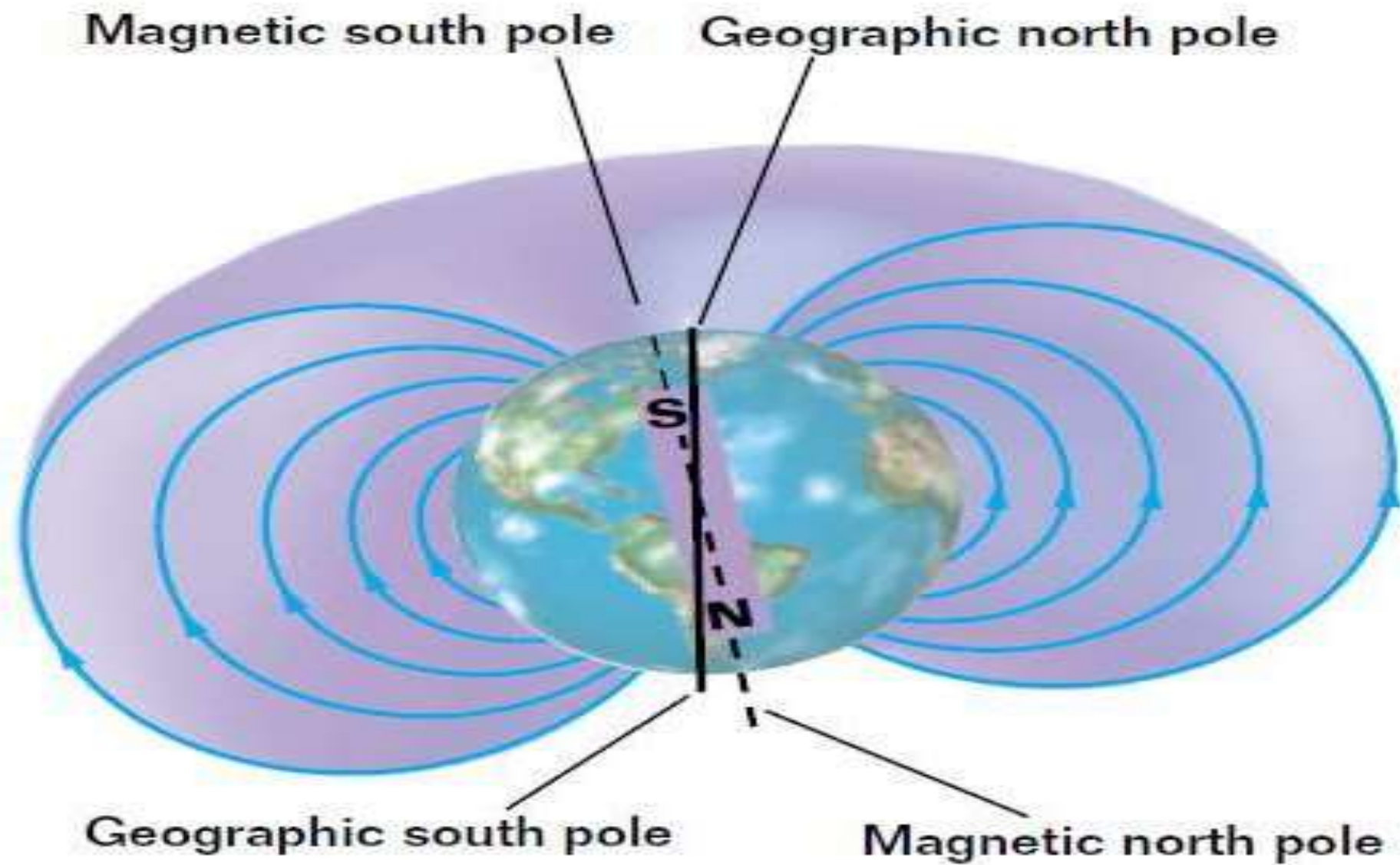
There are two important differences:

1. Electric field lines are in the direction of the electric force on a positive charge, but magnetic field lines are **perpendicular** to the magnetic force on a **moving** charge.
2. Electric field lines begin on positive charges and end on negative charges; magnetic field lines neither begin nor end.

Magnetic Field of the Earth

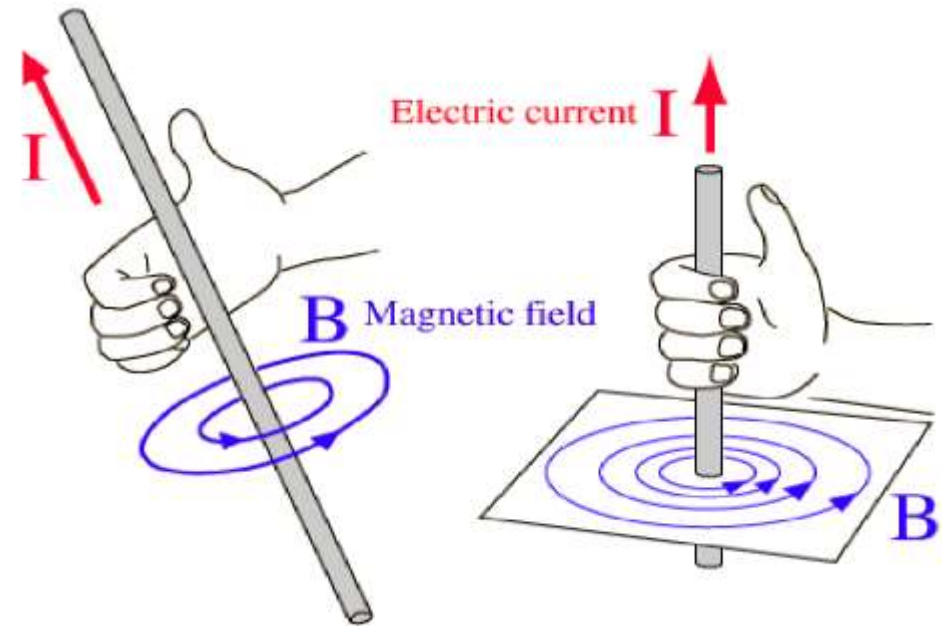
- The Earth's geographic North pole is actually the magnetic south pole
 - The north pole of a compass points towards geographic north and since opposites attract, we know that the Earth's geographic pole is magnetic south





Magnetic Field of a wire

- The “Right Hand Rule” for the magnetic field
- Point your thumb in the direction of the current and curl your fingers in the direction of the field



Magnetic Force

- A charge moving through a magnetic field experiences a force

$$F_{\text{magnetic}} = qvB$$

q = magnitude of charge

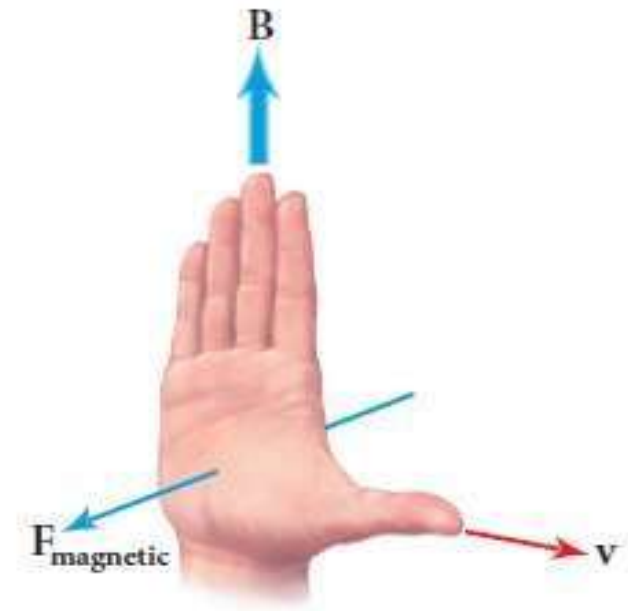
v = speed of charge

B = Strength of the magnetic field (measured in Tesla, T)

A second Right-Hand Rule

A second Right-Hand Rule

- Of course, force is a vector!
- To find the direction of the magnetic force use another right hand rule
 - Fingers point in direction of the field
 - Thumb points in direction of v
 - Palm points in direction of magnetic force



Conventions for direction of field

Direction of Field	Symbol
Into the page	X
Out of the page	•

WARNING: The right-hand rule is for the direction of the force acting on a POSITIVE CHARGE.

To find the direction of the force acting on a negative charge, you'll have to use the rule and change the sign!

Sample Problem p. 775 #2 (edited)

- A proton traveling to the right along the x-axis enters a region where there is a magnetic field of 2.5 T directed north. If the proton experiences a force of 3.2×10^{-12} N, find the speed of the proton. What is the direction of the force exerted on the proton?

The speed of the proton

$$v = \frac{F_{\text{magnetic}}}{qB} = \frac{3.2 \times 10^{-12} \text{ N}}{(1.6 \times 10^{-19} \text{ C})(2.5 \text{ T})} = 8.0 \times 10^6 \frac{\text{m}}{\text{s}}$$

- What's the direction of F? Use the RHR!!
 - v is east, B is north...F is....
 - Out of the page!
 - If it was an electron, the force would be into the page!

Sample Problem (not in book)

Sample Problem (not in book)

- An electron is moving with a velocity of 6×10^6 m/s westward in a 3.0 T magnetic field that is pointed out of the page.
- Find the magnitude and direction of the force acting on the electron.

Sample Problem

$$F_{\text{magnetic}} = qvB = (1.6 \times 10^{-19} \text{ C})(6 \times 10^6 \frac{\text{m}}{\text{s}})(3.0 \text{ T})$$

- $F = 2.88 \times 10^{-12} \text{ N}$
- Direction? Use the RHR
- F points SOUTH (remember it's an electron!!)

Lorentz force

- The Lorentz force is the force exerted on a charged particle moving in electric and magnetic fields.

- $$\mathbf{F} = q\mathbf{\bar{E}} + q\mathbf{\bar{v}}\mathbf{\bar{B}} = q(\mathbf{\bar{E}} + \mathbf{\bar{v}}\mathbf{\bar{B}})$$

- where E = Electric field intensity

Magnetic Fields (with Word Illustrations)

- **Imagine This:**

Think of magnetic field lines like invisible spaghetti noodles wrapping around a wire. When current flows, these noodles swirl around it in perfect loops. The more current you push through the wire, the thicker and stronger these noodles become.

☐ **Bar Magnet Analogy:**

If you sprinkle iron filings around a bar magnet, they line up in curvy loops—those loops show the magnetic field. The lines emerge from the North pole, loop around through the air, and dive back into the South pole.

LORENTZ FORCE

A charged particle moving in a magnetic field is like a car driving straight that suddenly hits a side wind pushing it off course.

- The velocity of the particle is the car's movement.
- The magnetic field is the side wind.
- The force from the magnetic field pushes the car sideways — the path becomes circular or spiral, not straight!

- **Right-Hand Rule (Again, but for Lorentz Force):**

- Point your fingers in the direction of velocity
- Point your palm in the direction of magnetic field
- Your thumb shows the direction of the force on a positive charge.

- [?] Electron? Flip the direction, since electrons are negative!

MOTION OF CHARGED PARTICLES IN MAGNETIC FIELDS

- **MAGNETIC FIELDS INFLUENCE ONLY MOVING ELECTRIC CHARGES. UNLIKE ELECTRIC FIELDS, MAGNETIC FIELDS CANNOT CHANGE THE SPEED (KINETIC ENERGY) OF A CHARGED PARTICLE BECAUSE THE MAGNETIC FORCE IS ALWAYS PERPENDICULAR TO THE DIRECTION OF MOTION.**
- **INSTEAD, THE MAGNETIC FIELD CHANGES THE DIRECTION OF MOTION, PRODUCING CIRCULAR, HELICAL, OR SPIRAL TRAJECTORIES.**
- **THIS PRINCIPLE IS THE BASIS OF MANY MODERN TECHNOLOGIES, INCLUDING:**
 - **PARTICLE ACCELERATORS**
 - **CYCLOTRONS**
 - **MASS SPECTROMETERS**
 - **CATHODE RAY TUBES**
 - **MAGNETIC CONFINEMENT FUSION**
 - **ELECTRON MICROSCOPES**

MAGNETIC FORCE ON A MOVING CHARGE

A charged particle moving in a magnetic field experiences a magnetic force given by the Lorentz force law.

$$\vec{F} = q(\vec{v} \times \vec{B})$$

where

- F = magnetic force (N)
- q = electric charge (C)
- v = particle velocity (m/s)
- B = magnetic flux density (Tesla)
- $\times$ = vector cross product

Magnitude:

$$F = qvB \sin \theta$$

where

θ is the angle between velocity and magnetic field.

PROPERTIES OF MAGNETIC FORCE

The magnetic force is

- perpendicular to velocity
- perpendicular to magnetic field
- maximum when

$$\theta = 90^\circ$$

- zero when

$$\theta = 0^\circ$$

THE DIRECTION IS DETERMINED USING THE RIGHT-HAND RULE.

- **FOR POSITIVE CHARGE:**
- **POINT FINGERS ALONG THE VELOCITY.**
- **ROTATE FINGERS TOWARD THE MAGNETIC FIELD.**
- **THUMB POINTS ALONG THE FORCE.**
- **FOR NEGATIVE CHARGES (ELECTRONS), THE FORCE IS OPPOSITE.**

MOTION PARALLEL TO MAGNETIC FIELD

When

$$\theta = 0^\circ$$

then

$$F = qvB \sin 0 = 0$$

No magnetic force acts.

Therefore,

- speed remains constant
- direction remains constant
- particle moves in a straight line.

MOTION PERPENDICULAR TO MAGNETIC FIELD

Suppose

$$\theta = 90^\circ$$

Then

$$F = qvB$$

The force is always perpendicular to motion.

Hence it behaves as the centripetal force.

Therefore,

$$qvB = \frac{mv^2}{r}$$

Cancelling v ,

$$r = \frac{mv}{qB}$$

This is called the radius of curvature.

RADIUS OF CIRCULAR MOTION

RADIUS DEPENDS ON

- **PARTICLE MASS**
- **VELOCITY**
- **CHARGE**
- **MAGNETIC FIELD STRENGTH**
- **INCREASING VELOCITY INCREASES RADIUS.**
- **INCREASING MAGNETIC FIELD DECREASES RADIUS.**

ANGULAR VELOCITY

Angular velocity

$$\omega = \frac{v}{r}$$

Substitute

$$r = \frac{mv}{qB}$$

Therefore,

$$\omega = \frac{qB}{m}$$

CYCLOTRON FREQUENCY

- **CYCLOTRON FREQUENCY (ALSO CALLED THE GYROFREQUENCY) IS THE NUMBER OF COMPLETE CIRCULAR REVOLUTIONS PER SECOND MADE BY A CHARGED PARTICLE MOVING PERPENDICULAR TO A UNIFORM MAGNETIC FIELD.**
- **WHEN A CHARGED PARTICLE (SUCH AS AN ELECTRON OR PROTON) ENTERS A MAGNETIC FIELD AT RIGHT ANGLES, THE MAGNETIC FORCE ACTS AS THE CENTRIPETAL FORCE, CAUSING THE PARTICLE TO MOVE IN A CIRCULAR PATH. THE RATE AT WHICH THE PARTICLE COMPLETES ONE FULL REVOLUTION IS CALLED THE CYCLOTRON FREQUENCY.**

Frequency

$$f = \frac{\omega}{2\pi}$$

Hence

$$f = \frac{qB}{2\pi m}$$

Notice that frequency is independent of particle speed.

This principle is used in cyclotrons.

MOTION AT AN ANGLE TO MAGNETIC FIELD

Suppose velocity makes an angle θ .

Velocity has two components

Parallel component

$$v_{\parallel} = v \cos \theta$$

Perpendicular component

$$v_{\perp} = v \sin \theta$$

The perpendicular component produces circular motion.

The parallel component produces forward motion.

The result is a helical (spiral) path.

MOTION IN NON-UNIFORM MAGNETIC FIELDS

- **WHEN THE MAGNETIC FIELD CHANGES WITH POSITION,**
- **THE PARTICLE**
- **SPIRALS**
- **MIRRORS**
- **BECOMES TRAPPED**
- **THIS PHENOMENON OCCURS IN EARTH'S MAGNETOSPHERE.**
- **EXAMPLE:**
- **VAN ALLEN RADIATION BELTS**

COMBINED ELECTRIC AND MAGNETIC FIELDS

When both fields exist,
the total force becomes

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

This is the complete Lorentz force equation.

Applications include

- velocity selector
- mass spectrometer
- plasma confinement

ENERGY CONSIDERATIONS

Since magnetic force is perpendicular to displacement,

$$W = F \cdot d \cos 90^\circ$$

Thus,

$$W = 0$$

Therefore,

- magnetic fields do no work
- kinetic energy remains constant
- speed remains constant
- only direction changes

EXAMPLES

An electron enters a uniform magnetic field of 0.40 T perpendicular to the field with a speed of $4.0 \times 10^6 \text{ m/s}$

Calculate its radius of motion.

Given

- $m = 9.11 \times 10^{-31} \text{ kg}$
- $q = 1.6 \times 10^{-19} \text{ C}$

Solution

$$r = \frac{mv}{qB}$$

Substitute

$$r = \frac{(9.11 \times 10^{-31})(4.0 \times 10^6)}{(1.6 \times 10^{-19})(0.40)}$$
$$r = 5.69 \times 10^{-5} \text{ m}$$

TUTORIAL QUESTIONS

QUESTION 1

- A PROTON ENTERS A MAGNETIC FIELD OF 0.5 T PERPENDICULAR TO THE FIELD AT $2 \times 10^6 \text{ m/s}$. CALCULATE A. MAGNETIC FORCE, B. RADIUS, C. PERIOD

QUESTION 2

- AN ELECTRON ENTERS A MAGNETIC FIELD AT 45° , SHOW THAT ITS TRAJECTORY IS HELICAL, AND DETERMINE THE RADIUS AND PITCH.

- **A PROTON MOVES WITH A VELOCITY OF 3.5×10^6 M/S IN A MAGNETIC FIELD OF 0.25 T AT AN ANGLE OF 60° . CALCULATE: (A) THE MAGNETIC FORCE, (B) THE RADIUS OF THE HELICAL PATH, (C) THE PITCH OF THE HELIX.**

MODULE 4: MAGNETISM & EM INDUCTION

UNIT 2: SOURCES OF MAGNETIC FIELDS

MAGNETIC FORCE ON CURRENT-CARRYING CONDUCTORS

- **A CURRENT-CARRYING CONDUCTOR PLACED IN A MAGNETIC FIELD EXPERIENCES A FORCE BECAUSE THE MOVING ELECTRIC CHARGES (ELECTRONS) WITHIN THE CONDUCTOR INTERACT WITH THE EXTERNAL MAGNETIC FIELD.**
- **THIS PHENOMENON IS THE OPERATING PRINCIPLE OF MANY ELECTRICAL DEVICES, SUCH AS:**
 - **ELECTRIC MOTORS**
 - **LOUDSPEAKERS**
 - **GALVANOMETERS**
 - **MOVING-COIL METERS**
 - **RAIL GUNS**

When a conductor of length L carries a current I in a magnetic field B , each moving charge experiences the Lorentz force. The combined effect on all the charges produces a net force on the conductor.

The magnitude of the force is given by

$$F = BIL \sin \theta$$

where:

- F = magnetic force (N)
- B = magnetic flux density (Tesla)
- I = current (A)
- L = length of conductor within the magnetic field (m)
- θ = angle between the current and the magnetic field

SPECIAL CASES

(a) Conductor Parallel to the Magnetic Field

$$\theta = 0^\circ$$

$$F = BIL \sin 0 = 0$$

Result: No magnetic force acts.

(b) Conductor Perpendicular to the Magnetic Field

$$\theta = 90^\circ$$

$$F = BIL$$

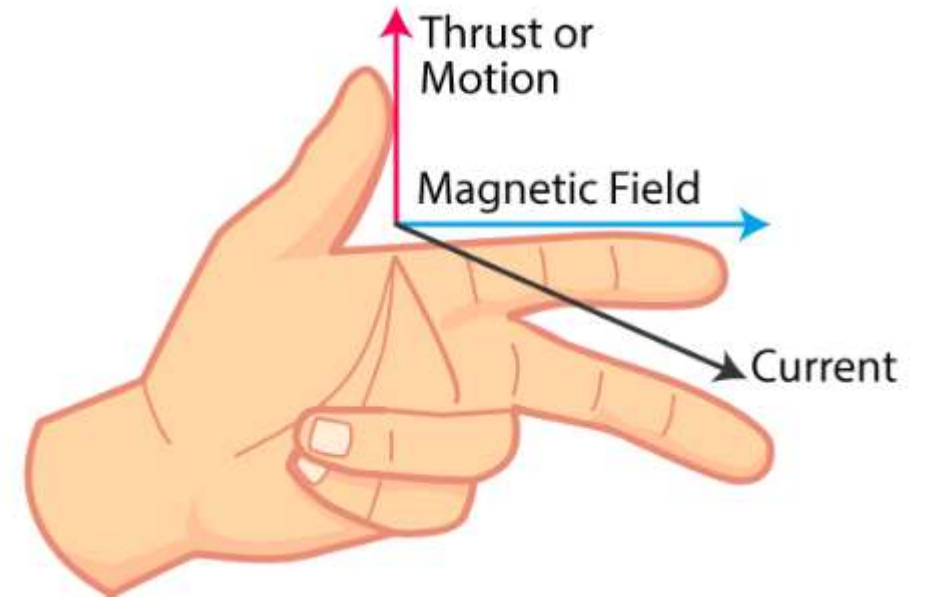
This is the maximum magnetic force.

DIRECTION OF THE MAGNETIC FORCE

THE DIRECTION OF THE FORCE IS DETERMINED USING FLEMING'S LEFT-HAND RULE.

HOLD THE LEFT HAND SO THAT:

- **FIRST FINGER → MAGNETIC FIELD (B)**
- **SECOND FINGER → CURRENT (I)**
- **THUMB → FORCE OR MOTION (F)**



FACTORS AFFECTING MAGNETIC FORCE

The force increases with:

- Magnetic field strength (B)
- Current (I)
- Length of conductor (L)
- Angle between current and field

It is maximum at

90°

and zero at

0°

BIOT–SAVART LAW

- **WHENEVER AN ELECTRIC CURRENT FLOWS THROUGH A CONDUCTOR, IT PRODUCES A MAGNETIC FIELD AROUND IT. THIS PHENOMENON WAS FIRST DISCOVERED IN 1820 BY THE DANISH PHYSICIST HANS CHRISTIAN ØRSTED, WHO OBSERVED THAT A COMPASS NEEDLE DEFLECTED WHEN PLACED NEAR A CURRENT-CARRYING WIRE.**
- **THE QUANTITATIVE RELATIONSHIP BETWEEN A CURRENT ELEMENT AND THE MAGNETIC FIELD IT PRODUCES WAS LATER DEVELOPED INDEPENDENTLY BY THE FRENCH PHYSICISTS JEAN-BAPTISTE BIOT AND FÉLIX SAVART IN 1820. THIS RELATIONSHIP IS KNOWN AS THE BIOT–SAVART LAW.**
- **THE LAW IS FUNDAMENTAL IN ELECTROMAGNETISM BECAUSE IT ALLOWS US TO CALCULATE THE MAGNETIC FIELD PRODUCED BY ANY CURRENT DISTRIBUTION.**

- **THE BIOT–SAVART LAW STATES THAT THE MAGNETIC FIELD AT A POINT DUE TO A SMALL CURRENT ELEMENT IS DIRECTLY PROPORTIONAL TO THE CURRENT AND THE LENGTH OF THE CURRENT ELEMENT, DIRECTLY PROPORTIONAL TO THE SINE OF THE ANGLE BETWEEN THE CURRENT ELEMENT AND THE LINE JOINING THE ELEMENT TO THE POINT, AND INVERSELY PROPORTIONAL TO THE SQUARE OF THE DISTANCE BETWEEN THEM.**

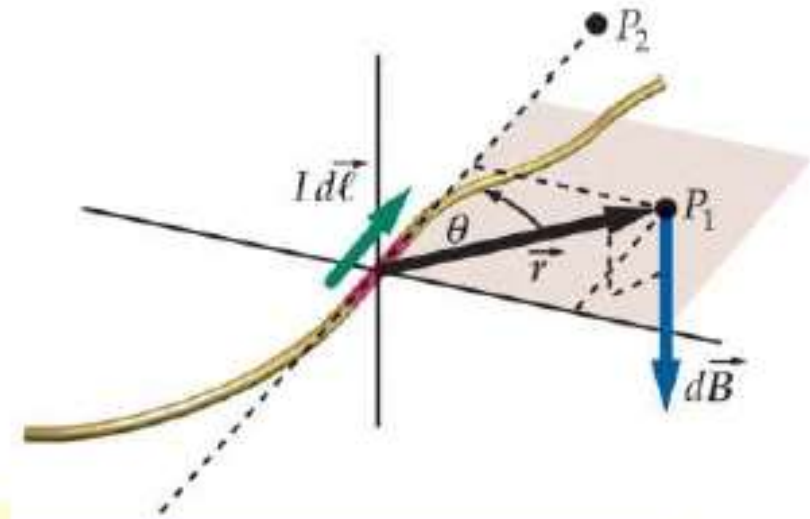
Board work

Biot-Savart law

The magnetic field $d\vec{B}$ produced by the current element $I d\vec{\ell}$ is given by

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \hat{r}}{r^2}$$

This is analogous to Coulomb's law for the electric field of a point charge.



However, note that the direction of $d\vec{B}$ is perpendicular to both $\underline{r}$ and $d\underline{\ell}$.

At point P_2 along the line of the current element, $d\vec{B}$ due to that element is zero.

The **Biot-Savart Law** relates the magnetic field generated by a small current element to the distance and direction from the element.

Where;

$d\vec{B}$ = magnetic field due to a small segment

μ_0 = Permeability of free space = $4\pi \times 10^{-7}$

I = current (A)

$d\vec{\ell}$ = length element vector in direction of current

$\hat{r}$ = unit vector from source to field point

r = distance between the source and point of interest

$$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{I d\vec{\ell} \times \hat{r}}{r^2}$$

Application: Straight Wire

The magnetic field a distance r from a long straight wire carrying current I :

$$B = \frac{\mu_0 I}{2\pi r}$$

MAGNETIC FIELD AT THE CENTRE OF A CIRCULAR LOOP

For a circular loop of radius R ,

$$B = \frac{\mu_0 I}{2R}$$

For a coil of N turns,

$$B = \frac{\mu_0 N I}{2R}$$

APPLICATIONS OF THE BIOT–SAVART LAW

THE BIOT–SAVART LAW IS WIDELY USED IN:

- **DESIGNING ELECTROMAGNETS.**
- **CALCULATING MAGNETIC FIELDS IN ELECTRIC MOTORS.**
- **MAGNETIC RESONANCE IMAGING (MRI).**
- **PARTICLE ACCELERATORS.**
- **MAGNETIC SENSORS.**
- **TRANSFORMERS.**
- **INDUCTORS.**
- **SOLENOIDS.**
- **ELECTRICAL TRANSMISSION SYSTEMS.**

EXAMPLE

- **A LONG STRAIGHT CONDUCTOR CARRIES A CURRENT OF 12 A. CALCULATE THE MAGNETIC FIELD AT A POINT 5 CM FROM THE CONDUCTOR.**

Using

$$B = \frac{\mu_0 I}{2\pi r}$$

Substitute

$$B = \frac{(4\pi \times 10^{-7})(12)}{2\pi(0.05)}$$

$$B = 4.8 \times 10^{-5} \text{ T}$$

AMPÈRE'S LAW

- **WHENEVER AN ELECTRIC CURRENT FLOWS THROUGH A CONDUCTOR, IT CREATES A MAGNETIC FIELD AROUND IT. WHILE THE BIOT–SAVART LAW CAN CALCULATE THE MAGNETIC FIELD PRODUCED BY ANY CURRENT DISTRIBUTION, THE CALCULATIONS MAY BECOME COMPLICATED FOR COMPLEX GEOMETRIES.**
- **TO SIMPLIFY THESE CALCULATIONS FOR SYSTEMS WITH HIGH SYMMETRY, THE FRENCH PHYSICIST ANDRÉ-MARIE AMPÈRE FORMULATED AMPÈRE'S CIRCUITAL LAW IN 1826.**
- **AMPÈRE'S LAW RELATES THE CIRCULATION OF THE MAGNETIC FIELD AROUND A CLOSED PATH TO THE TOTAL ELECTRIC CURRENT ENCLOSED BY THAT PATH.**
- **IT IS ONE OF THE FOUR FUNDAMENTAL EQUATIONS OF CLASSICAL ELECTROMAGNETISM KNOWN AS MAXWELL'S EQUATIONS.**

Ampère's Law states that:

The line integral of the magnetic field around any closed path is equal to the permeability of free space multiplied by the total current enclosed by the path.

The integral form of Ampère's Law is

$$\oint \mathbf{B} \cdot d\mathbf{\ell} = \mu_0 I_{\text{enc}}$$

where:

- $\mathbf{B}$ = magnetic field (Tesla)
- $d\mathbf{\ell}$ = infinitesimal length element along the closed path
- $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$ = permeability of free space
- I_{enc} = total current enclosed by the Amperian loop

The symbol



means integration around a closed path.

The quantity

$$\mathbf{B} \cdot d\mathbf{\ell}$$

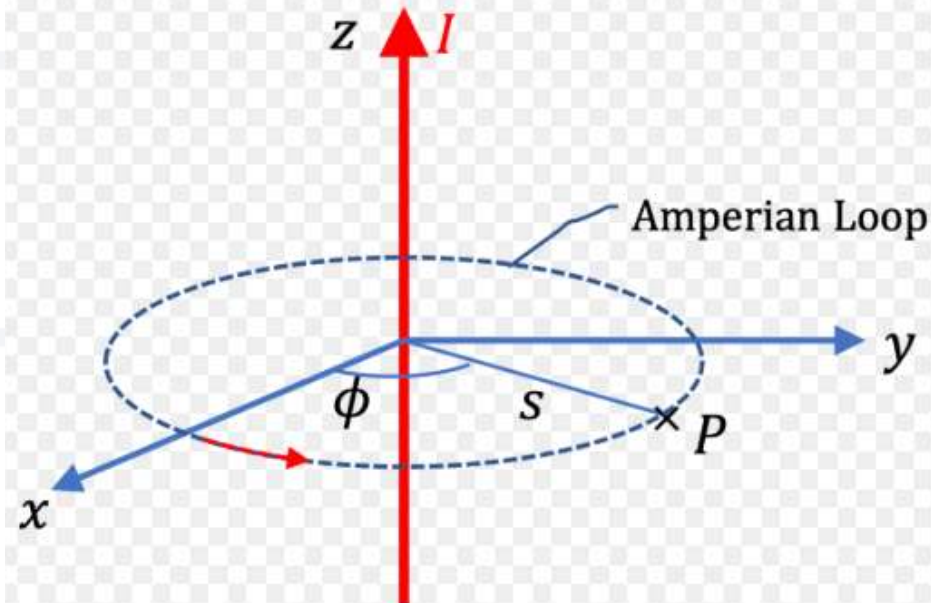
represents the component of the magnetic field tangent to the chosen path.

The law simply states that:

The stronger the enclosed current, the stronger the magnetic field around it.

THE AMPERIAN LOOP

- AN AMPERIAN LOOP IS AN IMAGINARY CLOSED PATH CHOSEN TO SIMPLIFY THE CALCULATION OF THE MAGNETIC FIELD.
- THE CHOICE OF THE LOOP DEPENDS ON THE SYMMETRY OF THE CURRENT DISTRIBUTION.



Consider a long straight conductor carrying current I .

Choose a circular Amperian loop of radius r .

Since B is constant on the loop,

$$\oint B \, dl = B \oint dl$$

The circumference of the loop is

$$2\pi r$$

Hence,

$$B(2\pi r) = \mu_0 I$$

Therefore,

$$B = \frac{\mu_0 I}{2\pi r}$$

INTERPRETATION

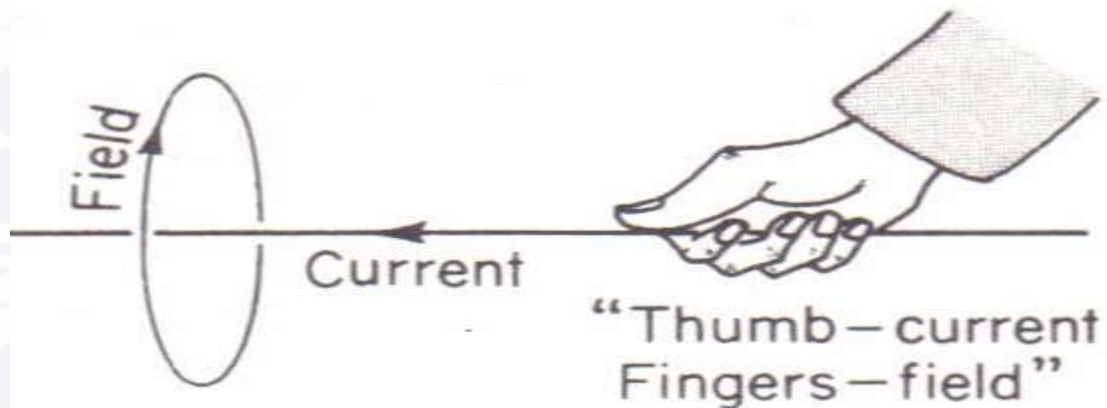
- **THE MAGNETIC FIELD**
- **INCREASES WITH CURRENT,**
- **DECREASES WITH DISTANCE FROM THE CONDUCTOR.**

DIRECTION OF MAGNETIC FIELD

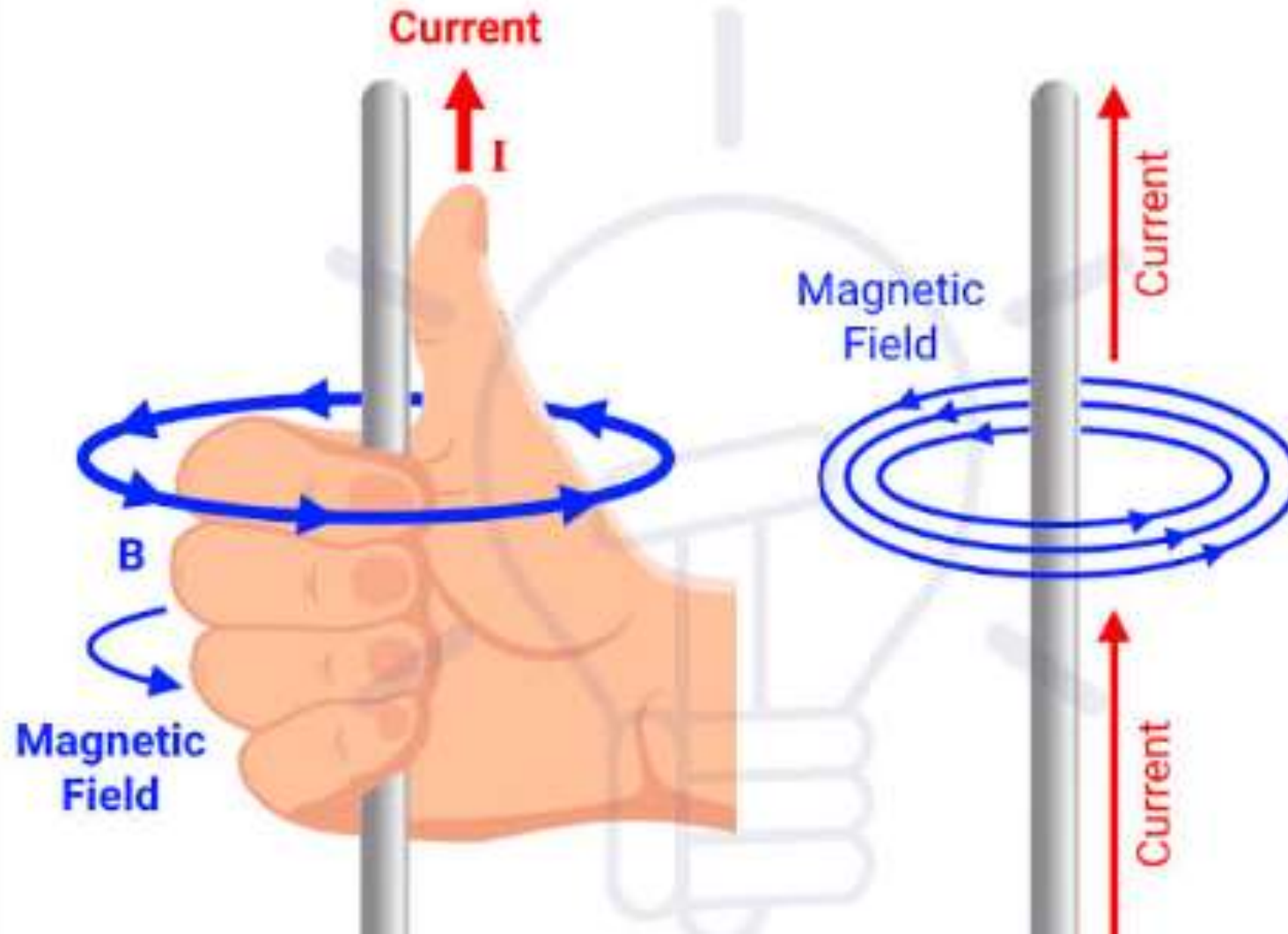
THE DIRECTION IS DETERMINED USING THE RIGHT-HAND GRIP RULE.

PROCEDURE

- **POINT THE THUMB IN THE DIRECTION OF CURRENT.**
- **CURL THE FINGERS.**
- **FINGERS INDICATE THE MAGNETIC FIELD DIRECTION.**



Right Hand Thumb/Grip Rule



MAGNETIC FIELD INSIDE A SOLENOID

A solenoid is a long coil of wire.

Inside a long solenoid,

- magnetic field is nearly uniform,
- field outside is very small.

Choose a rectangular Amperian loop.

Applying Ampère's Law,

$$BL = \mu_0 NI$$

where

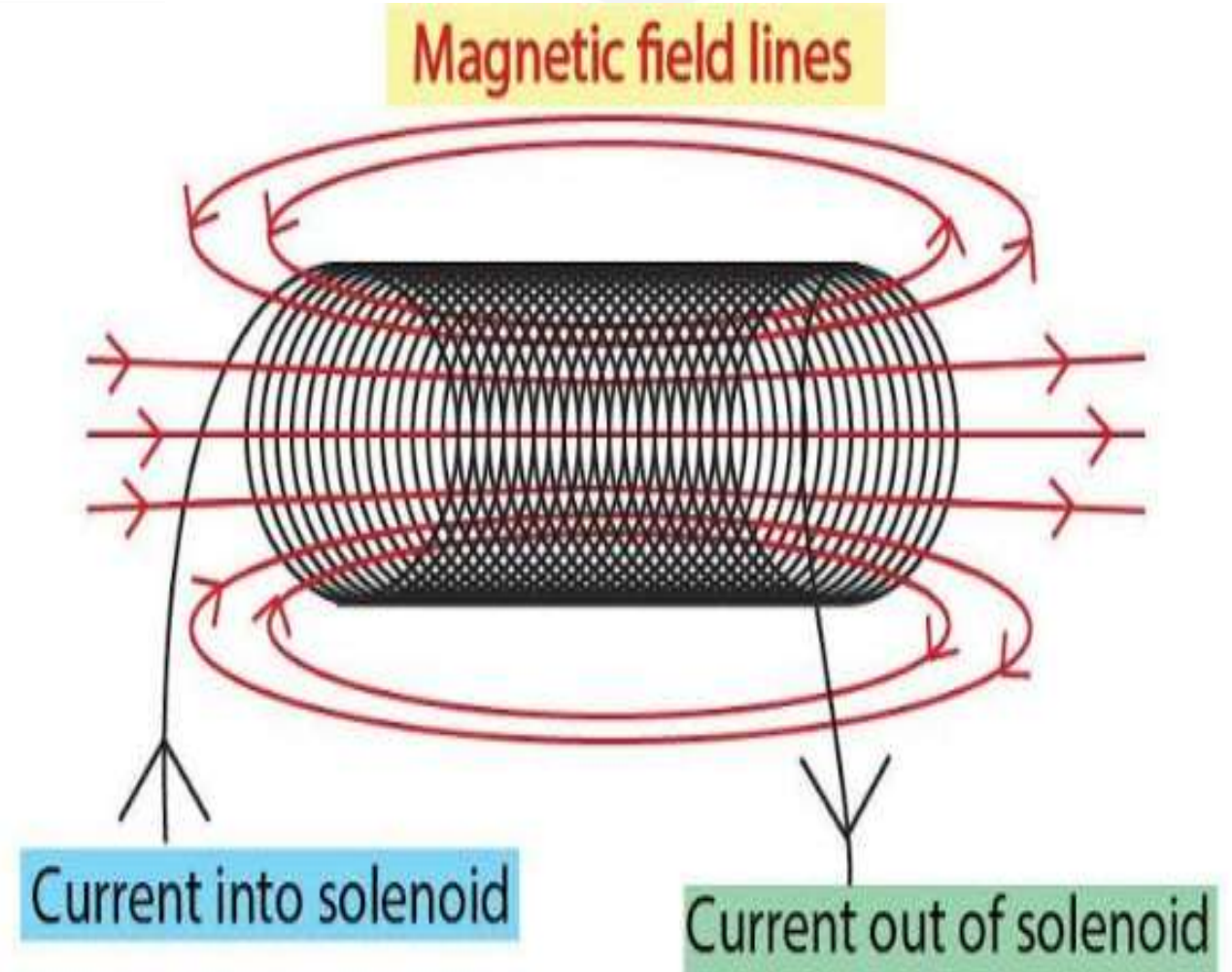
$$N = nL$$

Substituting,

$$B = \mu_0 nI$$

where

- n = number of turns per metre.



MAGNETIC FIELD INSIDE A TOROID

A toroid is a solenoid bent into a circular ring.

Using Ampère's Law,

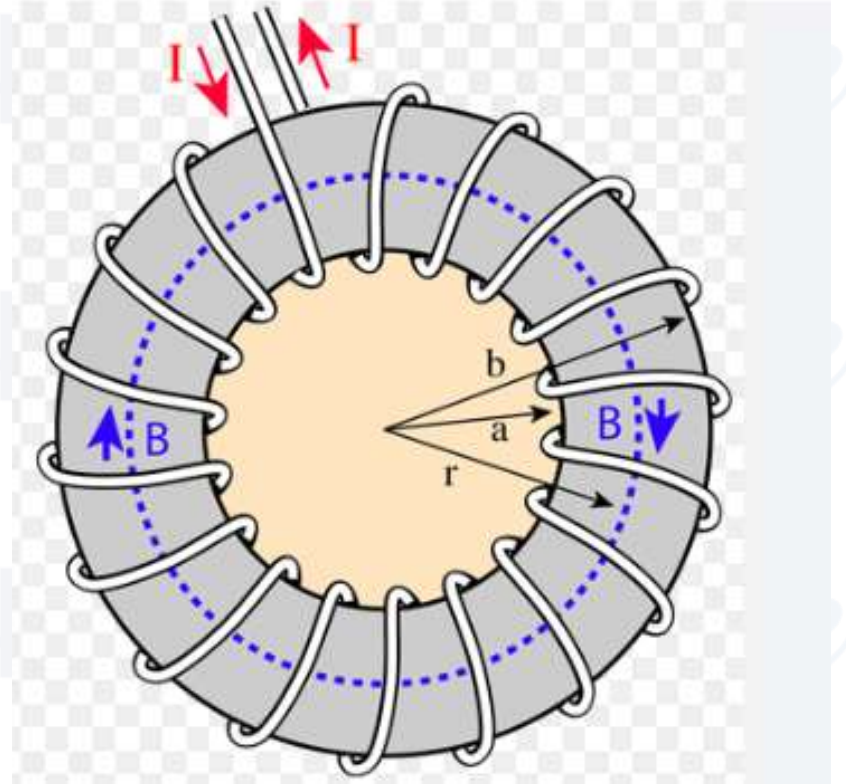
$$B(2\pi r) = \mu_0 N I$$

Hence,

$$B = \frac{\mu_0 N I}{2\pi r}$$

where

- N = number of turns.



DIFFERENTIAL FORM OF AMPÈRE'S LAW

One form of Maxwell's equations is

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

where

- $\mathbf{J}$ is current density.

AMPÈRE-MAXWELL LAW

For changing electric fields,

Maxwell modified Ampère's Law.

The complete equation becomes

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 \left(I + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

The additional term is called the displacement current.

APPLICATIONS

AMPÈRE'S LAW IS WIDELY USED IN

- **ELECTROMAGNET DESIGN**
- **ELECTRIC MOTORS**
- **TRANSFORMERS**
- **MRI SCANNERS**
- **SOLENOIDS**
- **TOROIDS**
- **INDUCTORS**
- **ELECTRICAL MACHINES**
- **POWER TRANSMISSION SYSTEMS**

EXAMPLE

- A STRAIGHT CONDUCTOR CARRIES 15 A. CALCULATE THE MAGNETIC FIELD 4 CM AWAY.

Given

$$I = 15A$$

$$r = 0.04m$$

Using

$$B = \frac{\mu_0 I}{2\pi r}$$

Substitute

$$B = \frac{4\pi \times 10^{-7} \times 15}{2\pi \times 0.04}$$

$$B = 7.5 \times 10^{-5}T$$

1. Magnetic Field around a Long Straight Wire

Q1: A straight wire carries a current of 4 A. Calculate the magnetic field 5 cm away from the wire.

Solution:

$$B = \frac{\mu_0 I}{2\pi r}$$
$$= \frac{4\pi \times 10^{-7} \times 4}{2\pi \times 0.05} = \frac{1.6 \times 10^{-6}}{0.1} = 1.6 \times 10^{-5} \text{ T}$$

2. Lorentz Force on a Moving Charge

Q2: A proton ($q = 1.6 \times 10^{-19} \text{ C}$) is moving at $3 \times 10^6 \text{ m/s}$ perpendicular to a magnetic field of strength 0.2 T . What is the magnitude of the force on it?

Solution:

$$F = qvB = (1.6 \times 10^{-19})(3 \times 10^6)(0.2) = 9.6 \times 10^{-14} \text{ N}$$



4. Radius of Circular Path of a Charged Particle

Q4: An electron of mass $9.1 \times 10^{-31} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$ enters a magnetic field of 0.01 T perpendicularly with velocity $2 \times 10^6 \text{ m/s}$. Find the radius of its circular path.

Solution:

$$F = qvB = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB}$$

$$r = \frac{9.1 \times 10^{-31} \times 2 \times 10^6}{1.6 \times 10^{-19} \times 0.01} = \frac{1.82 \times 10^{-24}}{1.6 \times 10^{-21}} = 1.14 \times 10^{-3} \text{ m or } 1.14 \text{ mm}$$

5. Magnetic Field Inside a Solenoid

Q5: A solenoid has 1000 turns per meter and carries 2 A. Find the magnetic field inside it.

Solution:

$$B = \mu_0 n I = 4\pi \times 10^{-7} \times 1000 \times 2 = 8\pi \times 10^{-4} \approx 2.51 \times 10^{-3} \text{ T}$$

9. Current from Ampère's Law

Q9: A circular loop of radius 20 cm encloses a magnetic field of $2 \times 10^{-6} \text{ T}$. What current flows through the loop?

$$B = \frac{\mu_0 I}{2\pi r} = I = \frac{2\pi r B}{\mu_0}$$



Plug in values:

$$I = \frac{2\pi \cdot 0.2 \cdot 2 \times 10^{-6}}{4\pi \times 10^{-7}}$$

Cancel π from numerator and denominator:

$$I = \frac{2 \cdot 0.2 \cdot 2 \times 10^{-6}}{4 \times 10^{-7}} = \frac{0.8 \times 10^{-6}}{4 \times 10^{-7}}$$

$$I = 2 \text{ A}$$

ANGLE BETWEEN VELOCITY AND MAGNETIC FIELD

A particle of charge $2 \times 10^{-6} \text{ C}$ moves at $4 \times 10^3 \text{ m/s}$ in a magnetic field of 0.2 T . If the angle between $\vec{v}$ and $\vec{B}$ is 30° , calculate the force.

Solution:

$$F = qvB \sin \theta$$

$$= (2 \times 10^{-6}) (4 \times 10^3) (0.2) (\sin 30^\circ)$$

$$= (1.6 \times 10^{-3}) (0.5) = 8.0 \times 10^{-4} \text{ N}$$

TUTORIAL QUESTIONS

A particle of charge $2 \times 10^{-6} \text{ C}$ moves at $4 \times 10^3 \text{ m/s}$ in a magnetic field of 0.2 T . If the angle between $\vec{v}$ and $\vec{B}$ is 30° , calculate the force.

An electron is moving parallel to a magnetic field of 0.5 T . What is the force on the electron?

An electron of mass $9.1 \times 10^{-31} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$ moves in a circular path of radius 0.05 m under the influence of a magnetic field. If its speed is $2 \times 10^6 \text{ m/s}$, calculate the magnetic field strength.

MODULE 4: MAGNETISM & EM INDUCTION

UNIT 3: ELECTROMAGNETIC INDUCTION

INTRODUCTION

- **ELECTROMAGNETIC INDUCTION IS THE PROCESS BY WHICH AN ELECTROMOTIVE FORCE (EMF) IS GENERATED IN A CONDUCTOR DUE TO A CHANGE IN THE MAGNETIC FIELD SURROUNDING IT.**
- **THIS PHENOMENON WAS DISCOVERED IN 1831 BY THE ENGLISH SCIENTIST MICHAEL FARADAY, WHO FOUND THAT A CHANGING MAGNETIC FIELD COULD INDUCE AN ELECTRIC CURRENT IN A NEARBY CONDUCTOR WITHOUT DIRECT ELECTRICAL CONTACT.**
- **ELECTROMAGNETIC INDUCTION IS THE FUNDAMENTAL PRINCIPLE BEHIND THE OPERATION OF:**
 - **ELECTRIC GENERATORS**
 - **TRANSFORMERS**
 - **INDUCTION MOTORS**
 - **WIRELESS CHARGING SYSTEMS**
 - **DYNAMIC MICROPHONES**
 - **METAL DETECTORS**
 - **INDUCTION COOKERS**

ELECTROMAGNETIC INDUCTION

- **ELECTROMAGNETIC INDUCTION IS THE PRODUCTION OF AN EMF (AND, IN A CLOSED CIRCUIT, AN ELECTRIC CURRENT) WHENEVER THE MAGNETIC FLUX LINKING A CONDUCTOR CHANGES.**

A CURRENT IS INDUCED ONLY IF:

- **THE MAGNETIC FIELD CHANGES.**
- **THE CONDUCTOR MOVES RELATIVE TO A MAGNETIC FIELD.**
- **THE AREA OF THE CONDUCTING LOOP CHANGES.**
- **THE ORIENTATION OF THE CONDUCTOR CHANGES.**

MAGNETIC FLUX

Magnetic flux measures the amount of magnetic field passing through a surface.

It is represented by the Greek letter Φ (phi).

The magnetic flux is given by

$$\Phi = BA \cos \theta$$

where:

- Φ = magnetic flux (Weber, Wb)
 - B = magnetic flux density (Tesla)
 - A = area (m^2)
 - θ = angle between the magnetic field and the normal to the surface
- The SI unit of magnetic flux is the Weber (Wb).

$$1 \text{ Wb} = 1 \text{ Tesla} \times 1 \text{ m}^2$$

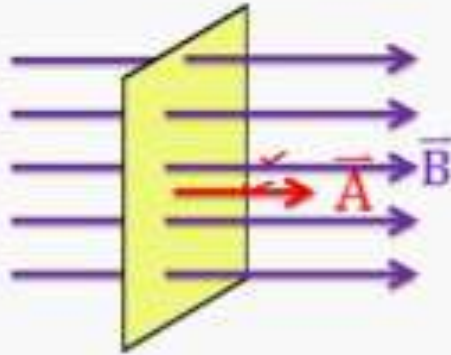
MAGNETIC FLUX AND FLUX DENSITY

"Magnetic flux Φ_B is the number of magnetic field lines passing through a certain area"

$$\Phi_B = \vec{B} \cdot \vec{A}$$

$$\Phi_B = BA \cos \theta$$

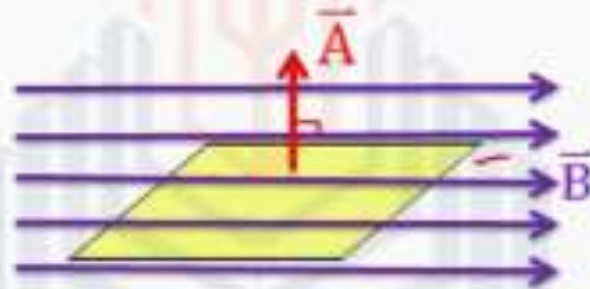
θ = angle between
vectors $\vec{B}$ and $\vec{A}$



$$\Phi_B = BA \cos 0^\circ$$

$$\Phi_B = BA$$

Maximum flux



$$\Phi_B = BA \cos 90^\circ$$

$$\Phi_B = 0$$

MAGNETIC FLUX LINKAGE

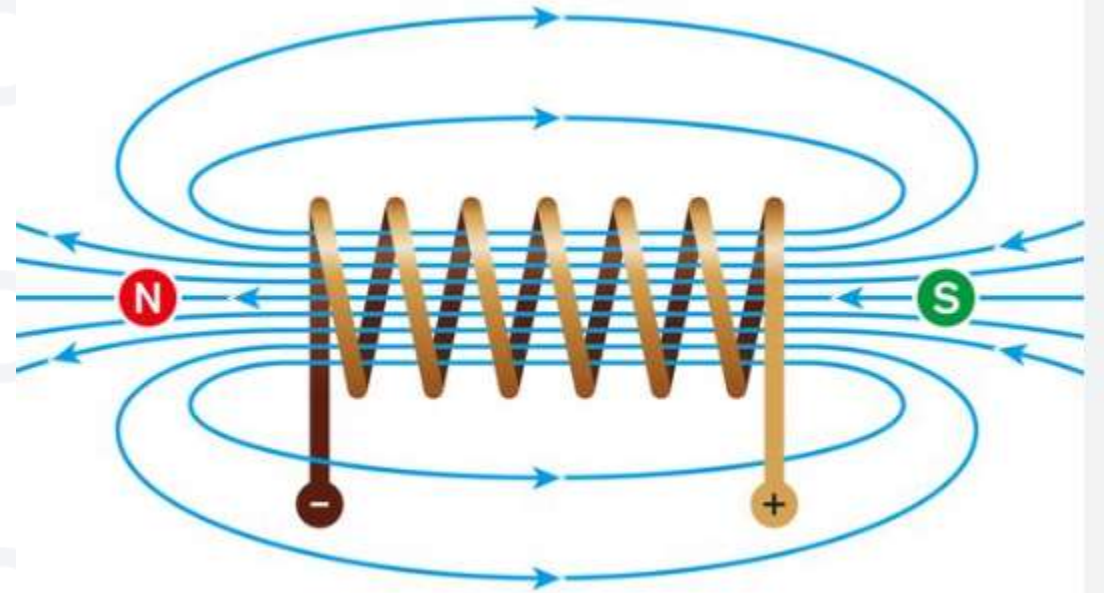
If a coil contains N turns,

$$\lambda = N\Phi$$

where

λ

is called flux linkage.



FARADAY'S FIRST LAW

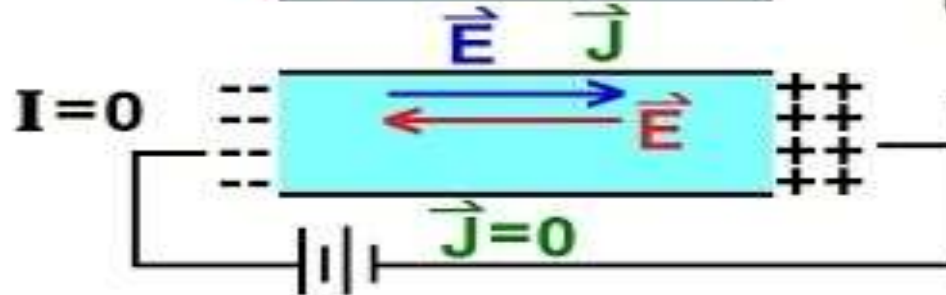
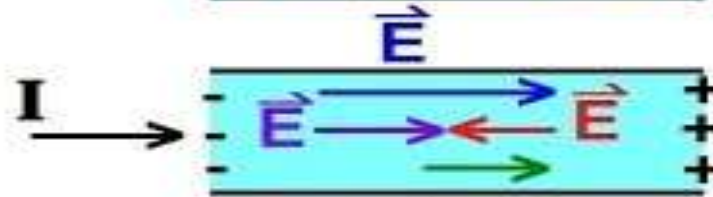
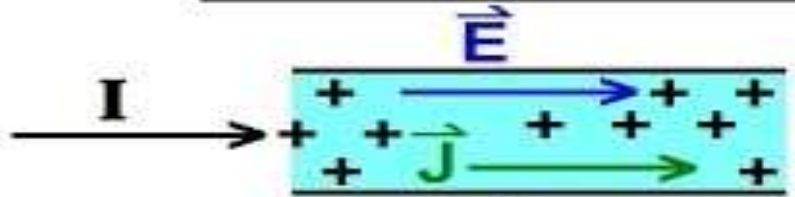
STATEMENT

WHENEVER THE MAGNETIC FLUX LINKING A CONDUCTOR CHANGES, AN EMF IS INDUCED IN THE CONDUCTOR.

- **IF THE CIRCUIT IS CLOSED,**
- **AN INDUCED CURRENT FLOWS.**

ELECTROMOTIVE FORCE

What is the Electromotive Force?

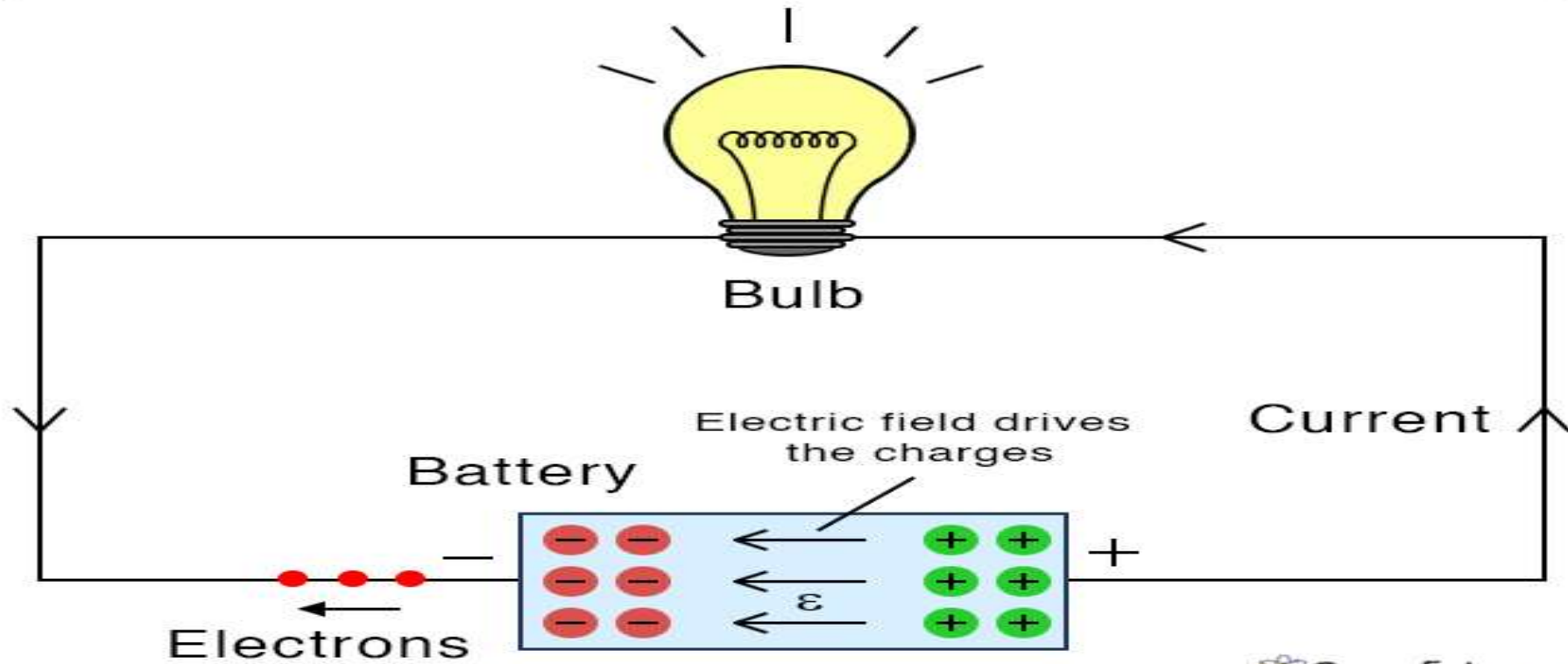


To keep the current flowing through the conductor there must be a force pushing the charges through the conductor.

The electromotive force (EMF)

ELECTROMOTIVE FORCE

Electromotive Force or EMF (ϵ)



ELECTROMOTIVE FORCE

ELECTROMOTIVE FORCE, ENERGY PER UNIT ELECTRIC CHARGE THAT IS IMPARTED BY AN ENERGY SOURCE, SUCH AS AN ELECTRIC GENERATOR OR A BATTERY.

- **ENERGY IS CONVERTED FROM ONE FORM TO ANOTHER IN THE GENERATOR OR BATTERY AS THE DEVICE DOES WORK ON THE ELECTRIC CHARGE BEING TRANSFERRED WITHIN ITSELF.**
- **ONE TERMINAL OF THE DEVICE BECOMES POSITIVELY CHARGED, THE OTHER BECOMES NEGATIVELY CHARGED.**
- **THE WORK DONE ON A UNIT OF ELECTRIC CHARGE, OR THE ENERGY THEREBY GAINED PER UNIT ELECTRIC CHARGE, IS THE ELECTROMOTIVE FORCE.**
- **ELECTROMOTIVE FORCE IS THE CHARACTERISTIC OF ANY ENERGY SOURCE CAPABLE OF DRIVING ELECTRIC CHARGE AROUND A CIRCUIT. IT IS ABBREVIATED E IN THE INTERNATIONAL METRIC SYSTEM, BUT ALSO POPULARLY, AS EMF.**

UNIT OF ELECTROMOTIVE FORCE

THE UNIT OF ELECTROMOTIVE FORCE IS THE VOLT (V). IF ONE COULOMB OF CHARGE IS DRIVEN THROUGH A POTENTIAL DIFFERENCE OF ONE VOLT, THE WORK DONE IS 1 JOULE.

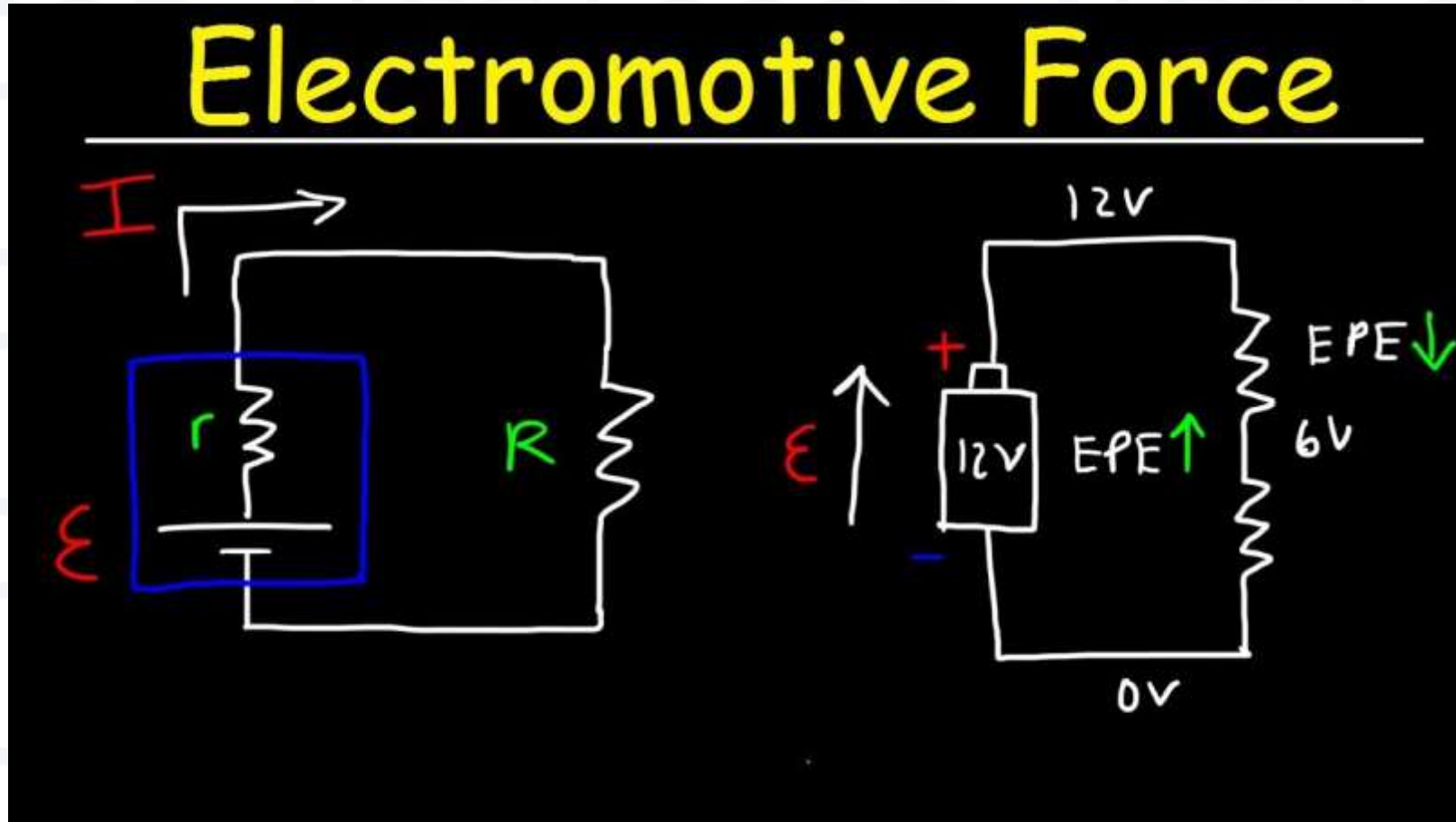
ELECTRICAL WORK IN JOULES IS THE PRODUCT OF CHARGE AND VOLTAGE.

DIFFERENCE BETWEEN VOLTAGE AND ELECTROMOTIVE FORCE?

ELECTROMOTIVE FORCE (EMF) IS THE POTENTIAL DIFFERENCE OF A SOURCE WHEN NO CURRENT IS FLOWING. TERMINAL VOLTAGE IS THE VOLTAGE OUTPUT OF A DEVICE MEASURED ACROSS IT'S TERMINALS.

**ELECTRIC POTENTIAL DIFFERENCE CREATES AN ELECTRIC FIELD THAT EXERTS FORCE
ON CHARGES CAUSING CURRENT FLOW**

ELECTROMOTIVE FORCE, INTERNAL RESISTANCE AND TERMINAL VOLTAGE



WORKED EXAMPLE

IN A CIRCUIT, THE EXTERNAL RESISTOR HAS VALUE 5Ω . THE INTERNAL RESISTANCE OF THE CELL IS 0.5Ω . IF THE CURRENT FLOWING IS 1 A . CALCULATE THE EMF OF THE CELL.

SOLUTION

USING $I = \frac{E}{R+r}$.

HERE E IS THE ELECTROMOTIVE FORCE, R IS THE EXTERNAL RESISTANCE, r IS THE INTERNAL RESISTANCE AND I IS THE CURRENT

SOLUTION TO EXAMPLE 1

WE HAVE

$$1 = \frac{E}{5+0.5}$$

E = 5.5VOLTS

EXAMPLE 2

A CELL OF UNKNOWN EMF E AND INTERNAL RESISTANCE 2Ω IS CONNECTED TO A

5Ω RESISTANCE. IF THE TERMINAL P.D. V IS $1V$, CALCULATE E
SOLUTION

TERMINAL PD $= V = 1V$. THIS IS THE PD ACROSS THE 5Ω WIRE

THEREFORE CURRENT IN 5Ω WIRE $= \frac{V}{R} = \frac{1}{5} = 0.2A$

PD ACROSS INTERNAL RESISTANCE R OF $2\Omega = IR = 0.2 \times 2 = 0.4V$

SOLUTION TO EXAMPLE 2 CONTD

**EMF E = PD ACROSS EXTERNAL RESISTOR R + PD ACROSS
INTERNAL RESISTANCE r**

$$= 1 + 0.4 = 1.4V$$

ALTERNATIVE SOLUTION TO EXAMPLE 2

THE CURRENT FLOWING $I = \frac{V}{R} = \frac{1}{5} = 0.2A$

Now $I = \frac{E}{R+r} = \frac{E}{5+2} = \frac{E}{7}$

So $E = I \times 7 = 0.2 \times 7 = 1.4volts$

CLASS DRILL

A WIRE OF 4Ω IS CONNECTED TO A BATTERY OF EMF 6V AND INTERNAL RESISTANCE R. THE CURRENT FLOWING IS 1 A. CALCULATE R

EXAMPLE 3

- THE TERMINAL PD OF A BATTERY IS 12V, WHEN AN EXTERNAL RESISTANCE OF
- 20Ω IS CONNECTED AND 13.5V WHEN AN EXTERNAL RESISTANCE OF 45Ω
- IS CONNECTED. CALCULATE THE EMF OF THE BATTERY

SOLUTION TO EXAMPLE 3

FOR THE WHOLE CIRCUIT $I = \frac{E}{R+r}$

**FOR TERMINAL PD OR PD ACROSS EXTERNAL RESISTANCE
R**

$$I = \frac{V}{R} = \frac{12}{20} = \frac{E}{20+r} \dots\dots\dots(1)$$

$$\text{AND } I = \frac{V}{R} = \frac{13.5}{45} = \frac{E}{45+r} \dots\dots\dots(2)$$

$$\text{FROM (1)} \quad E = \frac{12}{20} (20 + r) = 12 + \frac{3}{5}r$$

SOLUTION TO EXAMPLE 3 CONTD

FROM (2)

$$E = \frac{13.5}{45} (45 + r) = 13.5 + \frac{3}{10}r$$

So $12 + \frac{3}{5}r = 13.5 + \frac{3}{10}r$

THEREFORE $\frac{3}{5}r - \frac{3}{10}r = 13.5 - 12$

$$\frac{3}{10}r = 1.5$$
$$r = 5\Omega$$

SOLUTION TO EXAMPLE 3 CONTINUED

- **FROM (1)** $\frac{12}{20} = \frac{E}{20+5}$
- $E = 15V$

FARADAY'S SECOND LAW

Statement

The magnitude of the induced emf is directly proportional to the rate of change of magnetic flux linkage.

Mathematically,

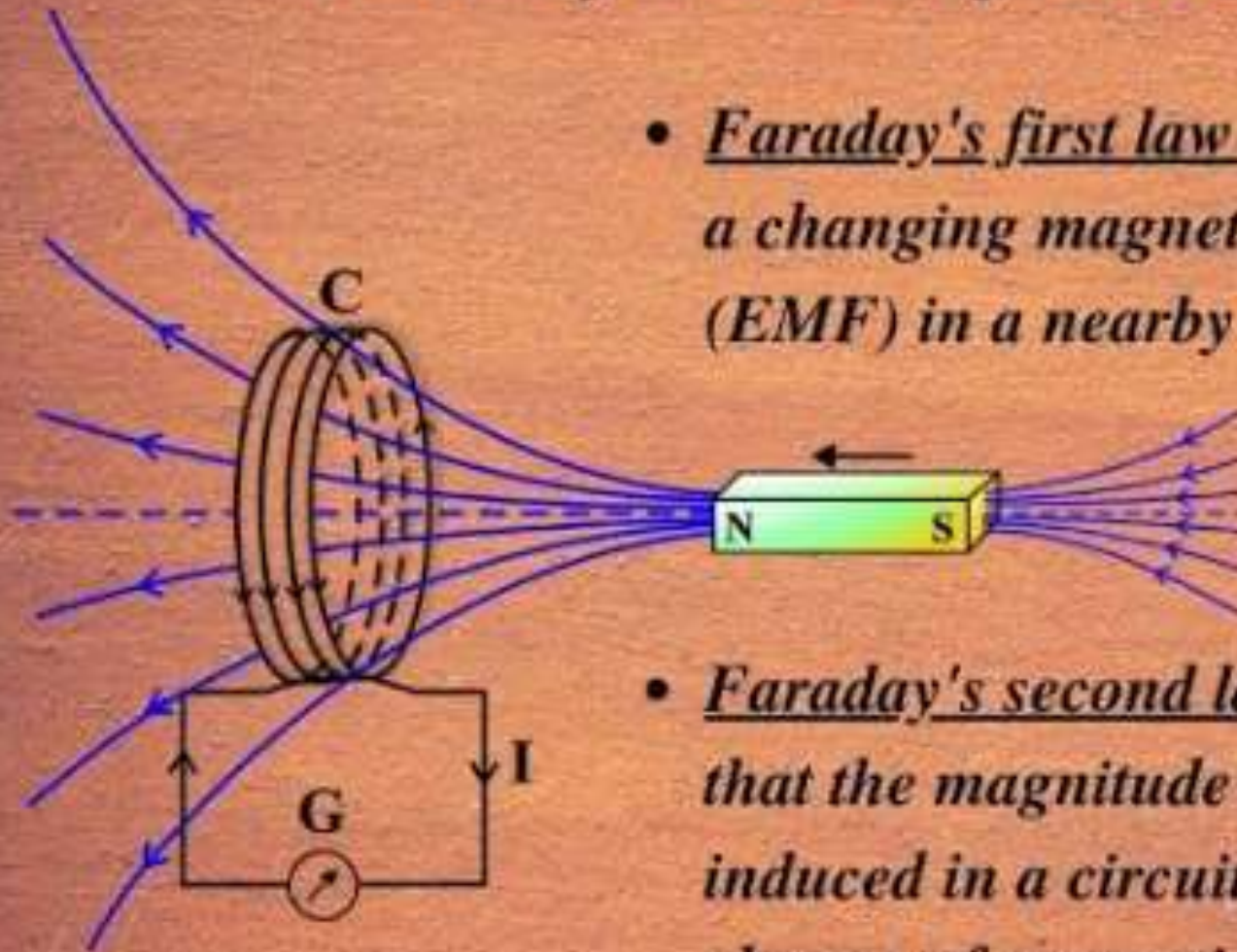
$$\varepsilon = -N \frac{d\Phi}{dt}$$

where

- ε = induced emf (V)
- N = number of turns
- Φ = magnetic flux (Wb)

The negative sign represents Lenz's Law.

Faraday's Laws of Electromagnetic Induction



- Faraday's first law of electromagnetic induction states that a changing magnetic field induces an electromotive force (EMF) in a nearby conductor.

$$\mathcal{E} = -\frac{d\phi}{dt}$$

- Faraday's second law of electromagnetic induction states that the magnitude of the electromotive force (EMF) induced in a circuit is directly proportional to the rate of change of magnetic flux through the circuit.

LENZ'S LAW

STATEMENT

- **THE DIRECTION OF THE INDUCED CURRENT IS SUCH THAT IT OPPOSES THE CHANGE IN MAGNETIC FLUX THAT PRODUCED IT.**
- **LENZ'S LAW IS A CONSEQUENCE OF THE LAW OF CONSERVATION OF ENERGY.**

MOTIONAL EMF

When a conductor moves through a magnetic field,

$$\varepsilon = BLv$$

where

- B = magnetic field
- L = length
- v = velocity

INDUCED CURRENT

If the conductor forms a closed circuit,

Ohm's Law gives

$$I = \frac{\mathcal{E}}{R}$$

EDDY CURRENTS

When a solid conductor experiences changing magnetic flux, closed loops of current are produced.

These are called eddy currents.

Applications include

- Magnetic braking
- Induction heating
- Metal detectors
- Speedometers

SELF-INDUCTION

- WHEN CURRENT CHANGES IN THE SAME COIL,
- AN EMF IS INDUCED IN THAT COIL.

$$\varepsilon = -L \frac{dI}{dt}$$

where

- L =inductance.

MUTUAL INDUCTION

- **WHEN CURRENT CHANGES IN ONE COIL, IT INDUCES AN EMF IN ANOTHER NEARBY COIL.**
- **THIS IS THE PRINCIPLE OF**
- **TRANSFORMERS**
- **WIRELESS CHARGING**
- **INDUCTION COOKERS**

TRANSFORMER

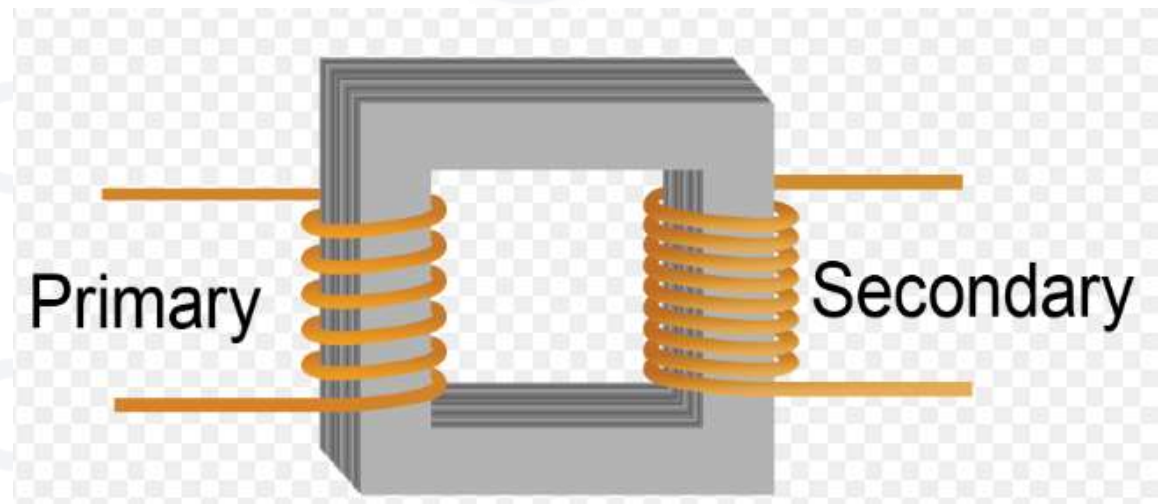
A transformer transfers electrical energy between two coils through mutual induction.

Transformer Equation

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

where:

- V_p = primary voltage
- V_s = secondary voltage
- N_p = primary turns
- N_s = secondary turns



APPLICATIONS OF ELECTROMAGNETIC INDUCTION

- **ELECTROMAGNETIC INDUCTION IS USED IN:**
- **ELECTRIC GENERATORS**
- **TRANSFORMERS**
- **INDUCTION MOTORS**
- **DYNAMIC MICROPHONES**
- **GUITAR PICKUPS**
- **WIRELESS CHARGING SYSTEMS**
- **INDUCTION COOKERS**
- **MAGNETIC BRAKES**
- **METAL DETECTORS**
- **ELECTRIC POWER TRANSMISSION**
- **MRI SYSTEMS**
- **ELECTRIC VEHICLE CHARGING**

EXAMPLE

- **A COIL OF 200 TURNS EXPERIENCES A CHANGE IN MAGNETIC FLUX FROM 0.08 Wb TO 0.02 Wb IN 0.10 s.**
- **CALCULATE THE INDUCED EMF.**

Using Faraday's Law,

$$\varepsilon = -N \frac{\Delta\Phi}{\Delta t}$$

Given:

- $N = 200$
- $\Delta\Phi = 0.08 - 0.02 = 0.06 \text{ Wb}$
- $\Delta t = 0.10 \text{ s}$

$$\begin{aligned}\varepsilon &= -\frac{200(0.06)}{0.10} \\ &= -120 \text{ V}\end{aligned}$$

The magnitude of the induced emf is

$$\boxed{120 \text{ V}}$$

(The negative sign indicates the direction of the induced emf as given by Lenz's Law.)